

MUHAMMAD RAMADIANSYAH

CTO, AI Tech Leader, AI, ML & Data Engineer

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Accomplished software developer with over 8 years of experience in senior machine learning engineering, data science, and full-stack web development. Seeking opportunities to apply expertise in cutting-edge technologies and data-driven solutions to drive innovation and deliver results.

PROFESSIONAL EXPERIENCE

INTURA PTE. LTD.

Singapore, Singapore

Co-Founder, CTO

Feb, 2025 - Present

- Develop Brand Research and AI Visibility Platform that evaluates brand visibility on AI platforms by generating and analyzing user queries
- Build Social Media Listening platform with dashboard and scraper pipelines for TikTok, Instagram, Threads, and X (Twitter)
- Build video and audio analysis engine
- Develop LLM Experimentation Platform and AI SDK for Intura LLM experimentation
- Set up Google Cloud Platform infrastructure (Compute Engine, Cloud SQL, Pub/Sub, BigQuery, Cloud Run, Cloud Build, Secret Manager)
- Set up internal tools spanning CRM, alerting and monitoring servers, and GenAI-powered finance tooling
- Develop monitoring and observability for LLM
- Deliver solutions for B2B clients across fashion, retail, SaaS, edutech, and beauty industries

PT. FINACCEL TEKNOLOGI INDONESIA

Jakarta, Indonesia

Senior Machine Learning Engineer

May, 2022 - Jun 2025

- Develop, maintain and monitor software applications and internal tools for machine learning solutions
- Optimize data pipeline process / end-to-end machine learning workflow / service processing cost
- Improve alert, monitoring, and architecture patterns for data services

Selected Project Experience

- AI Marketing Platform Project
 - Build web platform using React, Express and TypeScript combined with Python application
 - Deploy generative AI application for generating marketing copywriting and image banners
- Customer Relationship Management Project
 - Collaborate with Data Scientists for building product recommendation, propensity model, dynamic pricing and customer segmentation, increasing transaction rate around 5%–15% and decreasing churn rate around 4%–6%
 - Optimize cron services with big data approaches, cutting processing time by an average of 300%
 - Develop and maintain automatic A/B testing flow using Multi-Armed Bandit
 - Track and monitor 30+ AI/ML applications including CPU/memory usage, query usage and infra usage
- Operation, Data Platform and Internal Tools Project
 - Build, maintain and monitor computer vision solutions such as OCR and Face Anti-Spoofing that reduce operational cost
 - Build centralized 3rd-party data platform handling 300 requests per minute with average CPU/memory usage below 25%

- Build forecasting solutions for predicting Gross Marginal Value, run-out balance and revenues, preventing transaction failures by about 4%–8% and budget overruns capped at 20%

PT. ACCENTURE

Jakarta, Indonesia

AI Engineering Analyst

Oct, 2021- Apr, 2022

- Containerize chatbot services with a total of 90+ microservices
- Migrate 90+ microservices along with data storage and monitoring tools from on-premise infrastructure to AWS infrastructure
- Revamp using Kubernetes architecture, reducing latency around 50% at peak, decreasing error rate due to infrastructure issues from 9% to 6%, and handling 100–300 more requests per minute
- Integrate service discovery using Consul for AI chatbot services

PT. BHINNEKA MENTARI DIMENSI

Jakarta, Indonesia

AI Development Specialist

Jan, 2020- Oct, 2021

- Revamp and improve scraping engine from on-premise into Apache Airflow (Cloud Composer), Cloud SQL, Dataproc and BigQuery, reducing infrastructure cost per month by around 30M – 100M IDR
- Build Bulk Upload Tools to help new merchants register on the platform, contributing to 1%–3% merchant acquisitions
- Design, build and maintain data pipelines such as Content-Based Product Recommendation and e-commerce scraping

TEACHING AND FREELANCE EXPERIENCE

PT. HACKTIVATE TEKNOLOGI INDONESIA

Jakarta, Indonesia

Full Stack Web Development Instructor (Full Time)

Jun 2018 - Jun 2019

- Teach and assist students in back-end technology mainly using the Node.js stack (REST API, GraphQL, SQL and NoSQL)
- Teach and assist students in front-end technology (web and mobile) mainly using JavaScript libraries and frameworks (jQuery, Vue.js, React.js, React Native)

RAKAMIN ACADEMY

Jakarta, Indonesia

Head of Big Data and Machine Learning Instructor for Digital Bank (Part Time)

Apr 2023- May 2023

- Teach and assist with big data tools used in production such as Spark, Hadoop, Kafka and Airflow
- Build end-to-end machine learning solutions for chatbot
- Build end-to-end machine learning solutions for automating the KYC process such as KTP Reader
- Build solutions for machine learning monitoring and alerting in the production environment

RAKAMIN ACADEMY

Jakarta, Indonesia

Data Science Instructor (Part Time)

Apr 2022- Oct 2023

- Teach and assist with basic Python programming concepts and how to operate Git for collaboration
- Create and evaluate curriculum and videos for data processing modules such as filtering, sorting, feature engineering, removing duplicates and cleaning

JAKARTA LABS

Jakarta, Indonesia

AI Engineering Instructor (Part Time)

Apr 2022- Jun 2023

- Teach and assist students in machine learning deployment such as creating REST APIs for serving AI models, building event-driven systems and creating cron scripts for running machine learning tasks
- Teach and assist students to become familiar with production tools such as Docker, Kubernetes, Git and Cloud Computing

EDUCATION

UNIVERSITY OF INDONESIA

Depok, Indonesia

Master of Science, Major in Computer Science; Minor in Artificial Intelligence (GPA: 3.91)

2019-2021

- Generalized Proxy Anchor Loss for Seen Data and Unseen Data Face Verification (Thesis)
- Pair-based losses and Proxy-based losses for Face Image Retrieval:
 - Published on: [the 2020 International Conference on Advanced Computer Science and Information Systems \(ICACSIS\)](#)

UNIVERSITY OF INDONESIA

Depok Indonesia

Bachelor of Engineering, Major in Electrical Engineering; Minor in Control Systems (GPA: 3.36)

2013-2017

- Modeling and Analysis Servo Motor Control for Loading-Unloading (LUL) Robot Application on CNC Milling Machine (Thesis)
 - Published on: [The 15th Quality in Research \(QIR\), 2017](#)

ADDITIONAL TECHNICAL SKILLS

- **Programming Languages:** Python, JavaScript, TypeScript, Golang, Java
- **Machine Learning Modeling:** Forecasting, Regression, Classification, Clustering, Dimensionality Reduction, Similarity
- **Machine Learning Stack:** Prophet, Scikit-learn, K-Modes, CatBoost, LightGBM, Pandas, NumPy
- **Generative AI:** OpenAI API, Claude, Gemini, DALL-E and Vertex AI
- **Deep Learning Modeling:** CNN, RNN / LSTM / GRU, Transfer Learning, Metric Learning, Generative Networks, Auto Encoder
- **Deep Learning Stack:** OpenCV, PyTorch, NLTK, Gensim, Flair Embedding
- **Front End Development Stack:** React.js, Vue.js, HTML, CSS
- **Back End Development Stack:** FastAPI, Flask, Express, Rate Limiter, Circuit Breaker, Authentication
- **Data Storage Stack:** MySQL, PostgreSQL, MongoDB, Firebase, Redis
- **Big Data Stack:** Apache Spark, Apache Kafka, Apache Airflow, Apache Beam, HDFS
- **Google Cloud Platform Stack:** Cloud Composer, Pub/Sub, Cloud Run, BigQuery, Cloud SQL, Dataproc, Compute Engine, Cloud Function, Secret Manager, Cloud Storage
- **Amazon Web Services Stack:** ECS, EKS, EC2, Lambda, S3, RDS, ElastiCache, Parameter Store

AWARD AND CERTIFICATES

- **Best Session Presenter Award, Jakarta - the 12th ICACSIS 2020**
https://drive.google.com/file/d/19P0CmihedZokUMU7eei_uV9XbDie-QeK/view?usp=sharing
- **Presenter Participant, Jakarta - the 12th ICACSIS 2020**
<https://drive.google.com/file/d/1mTUhndOBliR-mIbCiCXgZf6pflkdzgo/view?usp=sharing>
- **Data Visualization and Machine Learning, Jakarta - Algoritma Data Science School**
https://drive.google.com/drive/u/1/folders/1mHK1daXQ8lIk1YJBkfdLnryj_TdNjhz4
- **Machine Learning Bootcamp, Jakarta - Evolve Machine Learners**
<https://drive.google.com/open?id=1FWeQx4hG3nd-JiEr6Tw0U9emi1AQ22f1>
- **Javascript Full Stack Developer Bootcamp, Jakarta - Hacktiv8**
<https://hire.hacktiv8.com/verify/fsjs/MuhammadRamadiansyah-20180511>